SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

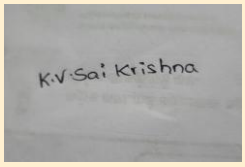
ASSESSMENT TOOL 2 – CASE STUDY ANALYSIS

Course Code	**CSA4315**	**Course Title**	**Internet Programming**
CO Assessed	**CO1**	**Assessment**	**Assessment Tool 2: Case Study**
Weightage	**30% of CIA**	**Total Marks**	**100**
CO2 Statement	Analyse, using a case study, the fundamental concepts of website development, the HTML5 document structure, semantic elements and the important features introduced in HTML5. (10 marks, CO1, BL3)		
Student Name	**K.V.SAI KRISHNA**	**Reg. No.**	**[192325037]**
Section/Batch	**[D / 2023]**	**Date**	**[31-10-2026]**

Code of Conduct:

I, K.V.SAI KRISHNA ([192325037]), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



CASE STUDY ANALYSIS

Question 1: Website Development and HTML5

1. Question Statement

Analyse, using a case study, the fundamental concepts of website development, the HTML5 document structure, semantic elements and the important features introduced in HTML5. (10 marks, CO1, BL3)

Approach followed. A case study website named **Campus Connect** (a college events portal) is designed and analysed. For each concept the report first explains the theory, then applies it to CampusConnect with working code, and finally interprets the result. Sections 2 and 3 cover website development fundamentals and the case study, Section 4 the document structure, Section 5 the semantic elements, Section 6 the new HTML5 features, and Sections 7 to 10 the analysis, summary and conclusion.

2. Introduction and Background

A website is a collection of linked web pages stored on a web server and identified by a domain name. When a user types a URL, the browser (the client) opens a TCP connection to the server (port 80 for HTTP, 443 for HTTPS), sends an HTTP request and receives an HTTP response that contains the page. The browser then reads the HTML, requests the linked CSS, JavaScript and image files, and renders the final page. In a LAMP or WAMP stack, Apache serves the files, PHP generates dynamic content and MySQL stores the data.

2.1 Fundamental concepts of website development

Concept	Explanation	Role in CampusConnect
Client–server model	Browser requests, server responds over HTTP/HTTPS	Student browser requests events.html from Apache
Front end	What the user sees and uses: HTML, CSS, JavaScript	Event listing, menus, registration form
Back end	Server-side logic and database: PHP, MySQL	Saves registrations, returns event data
Static and dynamic pages	Static pages are fixed files; dynamic pages are built per request	About page is static; event list comes from MySQL
Domain, hosting, URL	Name, server space and address of each resource	campusconnect.example.edu hosted on LAMP
Web standards (W3C, WHATWG)	Rules that make pages work in every browser	HTML5 living standard is followed
Responsive and accessible design	One site that adapts to all screens and all users	Works on phone, tablet and desktop

Table 1: Fundamental concepts of website development applied to the case study

2.2 The three layers of a web page

A modern page separates **structure, presentation and behaviour**. HTML defines the structure and meaning of the content, CSS defines the appearance (colour, layout, fonts) and JavaScript defines the behaviour (validation, animation, data requests). Keeping the three apart makes the site easier to maintain, because a change of colour touches only the CSS file and a change of content touches only the HTML.

2.3 Website development life cycle

Phase	Activity	CampusConnect example
1. Planning	Define purpose, users and scope	Publish college events and accept registrations

Phase	Activity	CampusConnect example
2. Design	Site map, wireframe, colour and fonts	Five pages and the layout in Figure 1
3. Development	Write HTML5, CSS3, JavaScript, PHP	Semantic pages, validated form, PHP handler
4. Testing	Validate code, test browsers and devices	W3C validator, mobile and desktop tests
5. Deployment	Upload to Apache server, configure domain	Copy files to /var/www/html
6. Maintenance	Fix errors, update content, monitor logs	Check access log status codes regularly

Table 2: Website development life cycle

3. The Case Study: CampusConnect Events Portal

Scenario. A university wants a public website where students can read about upcoming technical and cultural events, view photos and videos, and register online. An older version of the site was built with HTML 4 and generic tags. Search engines could not tell the navigation from the content, screen readers could not skip the menu, the registration form needed JavaScript libraries for simple checks, and video required a third-party plug-in. The task is to rebuild the site in HTML5.

No.	Requirement	HTML5 answer used
R1	Clear page structure understood by browsers, search engines and screen readers	Semantic elements: header, nav, main, section, article, aside, footer
R2	Event photos with captions	figure and figcaption
R3	Event promo video and audio	video and audio elements, no plug-in
R4	Online registration with checks	Form input types and validation attributes
R5	Seat map and chart of registrations	canvas and SVG
R6	Remember user choices, show nearby venues	Web Storage and Geolocation API
R7	Work on phones	Viewport meta tag and responsive CSS

Table 3: Requirements of the case study and the matching HTML5 features

3.1 Problem with the old HTML 4 structure

In the old site every region was a with a class such as "top", "menu" or "box". The names meant something only to the developer; the browser treated all of them as meaningless boxes. This style is often called "div soup". HTML5 replaces these boxes with elements whose names describe their purpose, so the meaning is built into the markup itself.

Aspect	HTML 4 / XHTML (old site)	HTML5 (new site)
Doctype	Long, version-specific DTD declaration	<!DOCTYPE html> (short, one line)
Character set	<meta http-equiv=...> long form	<meta charset="UTF-8">
Page regions	,	<header>, <nav>, <main>, <footer>
Video and audio	Flash or other plug-in	<video>, <audio> built into browser
Graphics	Images or Flash	<canvas> and inline <svg>
Form checks	JavaScript library needed	Built-in types and attributes
Local storage	Cookies only (4 KB)	localStorage, sessionStorage (about 5 MB)

Table 4: Comparison of the old and the new CampusConnect implementation

4. HTML5 Document Structure

Every HTML5 page follows the same skeleton. The **document type declaration** tells the browser to use standards mode. The **<html>** element is the root and carries the language of the page. The **<head>** holds information about the page that is not displayed (character set, title, viewport, style sheets), and the **<body>** holds everything the visitor sees.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <meta name="description" content="Events portal of the college">
  <title>CampusConnect | Home</title>
  <link rel="stylesheet" href="css/style.css">
</head>
<body>
  <!-- header, nav, main, aside, footer go here -->
  <script src="js/app.js" defer></script>
</body>
</html>
```

Listing 1: Basic HTML5 document skeleton of CampusConnect

Part	Purpose	Why it matters
<!DOCTYPE html>	Selects standards mode	Without it browsers use quirks mode and render pages inconsistently
<html lang="en">	Root element with page language	Helps screen readers choose pronunciation and search engines choose language
<meta charset="UTF-8">	Character encoding	Shows all characters correctly; must be within the first 1024 bytes
<meta name="viewport">	Controls width on mobile	Required for responsive layout on phones
<title>	Page title	Shown on the browser tab and in search results
<link> and <script>	Attach CSS and JavaScript	Keeps structure, style and behaviour separate; defer loads script after parsing
<body>	Visible content	Holds the semantic page structure

Table 5: Parts of the HTML5 document and their purpose

4.1 Syntax rules of HTML5

- Elements are written as a start tag, content and an end tag, for example text . Void elements such as <img>,
, <input> and <meta> have no end tag.
- Tags and attribute names are not case sensitive, but lower case is the accepted convention. Attribute values should be quoted.
- Elements must be nested properly: <article> ... </article> is valid while overlapping tags are not.
- File names matter on a LAMP server: Events.html and events.html are different files on Linux.

4.2 The document tree

The browser converts the markup into a tree called the **Document Object Model (DOM)**. The root is <html> with two children, <head> and <body>. JavaScript and CSS use this tree to find and change elements. A clean, semantic tree is therefore useful not only for reading but also for scripting and styling.

5. Semantic Elements in HTML5

A **semantic element** clearly describes its meaning to both the browser and the developer. and say nothing about their content, whereas <nav> says "this is a block of navigation links" and <article> says "this is an independent piece of content". HTML5 added a family of such elements to describe the common regions of a web page.

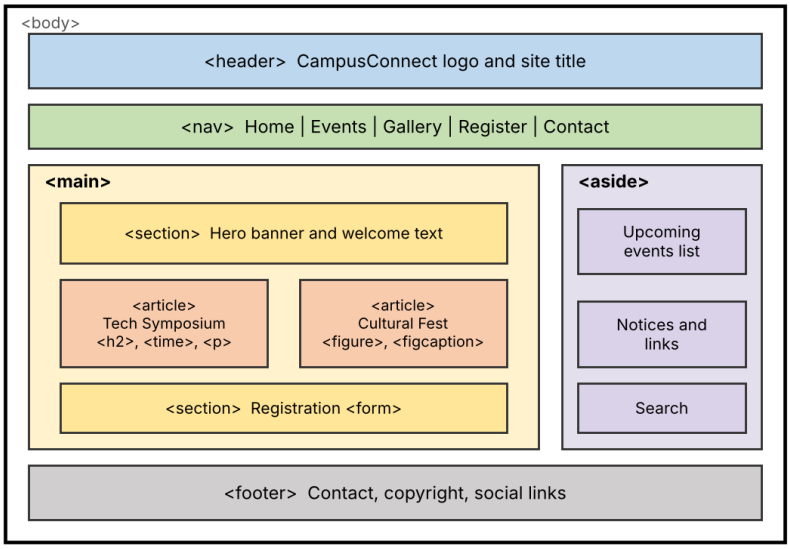


Figure 1: Semantic layout of the CampusConnect home page

Element	Meaning and use	Use in CampusConnect
<header>	Introductory content or navigation aids of a page or section	Logo, site title, tagline
<nav>	Major block of navigation links	Home, Events, Gallery, Register, Contact
<main>	Main unique content; only one per page	Everything between the menu and footer
<section>	Thematic group of content, normally with a heading	Hero banner, registration area
<article>	Self-contained content that can be reused on its own	Each event card
<aside>	Content related to, but separate from, the main content	Upcoming events, notices
<footer>	Footer of a page or section	Contact details, copyright
<figure>, <figcaption>	Illustration with a caption	Event photo and its caption
<time>	Machine-readable date and time	Event date: <time datetime="2026-12-05">
<mark>, <address>	Highlighted text; contact information	Search match; organiser contact

Table 6: Main semantic elements and their use in the case study

5.1 Semantic home page of CampusConnect

```
<body>
 <header>
  <h1>CampusConnect</h1>
  <p>Your college events in one place</p>
 </header>
 <nav aria-label="Main menu">
  <ul>
   <li><a href="index.html">Home</a></li>
   <li><a href="events.html">Events</a></li>
   <li><a href="register.html">Register</a></li>
  </ul>
 </nav>
 <main>
  <section>
   <h2>Upcoming Events</h2>
   <article>
    <h3>Tech Symposium 2026</h3>
    <p>Date: <time datetime="2026-12-05">5 December 2026</time></p>
    <figure>
     <img src="img/tech.jpg" alt="Students at the symposium">
     <figcaption>Last year's symposium</figcaption>
    </figure>
    <p>Talks, workshops and a coding contest.</p>
   </article>
  </section>
  <aside><h2>Notices</h2><p>Registration closes 30 November.</p></aside>
 </main>
 <footer><address>events@college.edu</address><p>&copy; 2026</p></footer>
</body>
```

Listing 2: Semantic body of the CampusConnect home page

5.2 Heading hierarchy and the document outline

The page has one <h1> for the site name, an <h2> for each section and an <h3> for each article. This hierarchy gives a clear outline. Screen-reader users move from heading to heading and from landmark to landmark (banner, navigation, main, complementary and content information), which the elements <header>, <nav>, <main>, <aside> and <footer> provide automatically without extra ARIA roles.

5.3 Choosing between section, article and div

Question to ask	If yes use	Example
Can the content stand alone and be shared or syndicated?	<article>	Event card, news item, blog post
Is it a themed group that would have a heading?	<section>	Upcoming events, registration
Is it only loosely related to the surrounding text?	<aside>	Notices, advertisements
Do you need a box only for styling or scripting?		Wrapper for a CSS grid

Table 7: Selecting the correct container element

6. Important Features Introduced in HTML5

6.1 Enhanced forms

HTML5 added new input types and attributes so that browsers can show suitable keyboards and date pickers and check the data without JavaScript. The registration form of CampusConnect uses them as follows.

```
<form action="register.php" method="post">
  <label for="n">Name</label>
  <input id="n" name="name" type="text" required minlength="3">
  <label for="e">Email</label>
  <input id="e" name="email" type="email" required placeholder="you@college.edu">
  <label for="m">Mobile</label>
  <input id="m" name="mobile" type="tel" pattern="[0-9]{10}" required>
  <label for="d">Preferred date</label>
  <input id="d" name="date" type="date" min="2026-11-10">
  <label for="ev">Event</label>
  <input id="ev" name="event" list="events">
  <datalist id="events"><option value="Tech Symposium"><option value="Cultural
Fest"></datalist>
  <button type="submit">Register</button>
</form>
```

Listing 3: HTML5 registration form with built-in validation

New input type or attribute	Function
email, url, tel, number, range, date, time, color, search	Specialised controls; mobile browsers show matching keyboard or picker
required, pattern, min, max, minlength	Browser blocks submission of invalid data and shows a message
placeholder, autofocus, autocomplete	Hint text, automatic focus, saved-value suggestions
<datalist>, <output>, <progress>, <meter>	Suggestion list, calculation result, task progress, gauge

Table 8: New form features

Note: client-side checks improve convenience only. The PHP script must validate the same data again on the server because a user can bypass the browser.

6.2 Native audio and video

The <video> and <audio> elements play media without Flash or other plug-ins. Several sources can be listed so that the browser picks a format it supports, and <track> adds subtitles for accessibility.

```
<video controls width="480" poster="img/poster.jpg" preload="metadata">
  <source src="media/promo.mp4" type="video/mp4">
  <source src="media/promo.webm" type="video/webm">
  <track kind="subtitles" src="media/promo-en.vtt" srclang="en" label="English">
  Your browser does not support video.
</video>
```

Listing 4: Event promo video with fallback formats and subtitles

6.3 Graphics: canvas and SVG

The <canvas> element gives a bitmap drawing area controlled by JavaScript, which suits charts, games and seat maps that change often. <svg> describes vector shapes in markup, so the graphics stay sharp at any size and each shape can be styled or made interactive. CampusConnect uses canvas for the live registration chart and SVG for the logo and icons.

```
<canvas id="seats" width="300" height="120"></canvas>
<script>
  const g = document.getElementById('seats').getContext('2d');
  g.fillStyle = '#2E75B6';
  g.fillRect(10, 20, 120, 80);   // booked seats
  g.fillStyle = '#C00000';
  g.fillRect(150, 20, 60, 80);   // remaining seats
</script>
<svg width="60" height="60"><circle cx="30" cy="30" r="25" fill="#2E75B6"/></svg>
```

Listing 5: Canvas and SVG graphics

6.4 JavaScript APIs

```
// Web Storage: remember the chosen event across visits
localStorage.setItem('lastEvent', 'Tech Symposium');
const last = localStorage.getItem('lastEvent');

// Geolocation: find the user's position for the venue map
navigator.geolocation.getCurrentPosition(
  pos => console.log(pos.coords.latitude, pos.coords.longitude),
  err => console.log('Location denied')
);
```

Listing 6: Web Storage and Geolocation

HTML5 feature	What it does	Used in CampusConnect for
Web Storage	Stores key-value data in the browser (about 5 MB), more than cookies and not sent with every request	Saving theme choice and last viewed event
Geolocation API	Reads device location, only with user permission	Showing route to the venue
Drag and Drop	Native dragging of elements	Reordering favourite events
Web Workers	Run JavaScript in a background thread	Heavy calculations without freezing the page
History API	Change the URL without reloading the page	Smooth navigation between event pages
Offline support (Service Worker, Cache)	Serve pages when the network fails	Reading the schedule offline
WebSocket	Two-way real-time connection	Live seat count
contenteditable, data-* attributes	Editable regions; custom data on elements	Organiser notes; event id on each card

Table 9: Other important HTML5 features

7. Analysis and Interpretation of the Case Study

7.1 Benefits obtained after moving to HTML5

Area	Observation in the old site	Improvement in the new site	Reason
Search engine optimisation	Crawler could not find the main content	Headings, article and time elements give clear signals	Meaningful tags describe content
Accessibility	Menu had to be read on every page	Landmarks let users skip to main	nav, main, aside are landmarks
Maintenance	Many nested divs with classes	Short readable markup	Tag names explain themselves
Page weight and speed	Plug-in and JavaScript libraries needed	Built-in video and validation	Fewer external files
Mobile use	Fixed width layout	Viewport tag and responsive CSS	Page adapts to screen
Development effort	Custom scripts for forms	Native input checks	Browser does the work

Table 10: Benefits of the HTML5 implementation (qualitative evaluation)

7.2 Validation and testing approach

- Check every page with the W3C Markup Validator to confirm the DOCTYPE, nesting and attributes.
- Test with a screen reader and keyboard only to confirm that landmarks and headings give a logical order.
- Test on Chrome, Firefox, Edge and Safari and on phone and desktop sizes, because support for some APIs differs.
- Use the browser developer tools to check the HTTP response of each resource: 200 for pages, and no 404 for missing images, CSS or scripts.

7.3 Best practices followed

- Use one <h1> and a logical heading order; do not choose headings for their size.
- Always give <img> an alt attribute and associate every form control with a <label>.
- Use and only when no semantic element fits.
- Keep CSS and JavaScript in separate files and load scripts with defer.
- Provide fallback text or formats for media, and validate again on the server.

7.4 Limitations

Older browsers may not support every feature (for example some input types or APIs), so fallbacks are needed. Geolocation and storage depend on user permission and privacy rules. Semantic tags give meaning but do not by themselves make a page accessible; text alternatives, contrast and keyboard access must still be provided. Finally, HTML5 is only the structure layer: styling needs CSS3 and server logic needs PHP and MySQL.

8. Summary of Answers

Part	Topic	Key points
1	Website development fundamentals	Client–server model, front end and back end, HTML/CSS/JS layers, development life cycle
2	HTML5 document structure	DOCTYPE, html with lang, head (charset, viewport, title), body, DOM tree
3	Semantic elements	header, nav, main, section, article, aside, footer, figure, time give meaning to the layout
4	New HTML5 features	Form types and validation, audio and video, canvas and SVG, Web Storage, Geolocation, Drag and Drop, Web Workers, History API, offline support
5	Case study result	CampusConnect became easier to maintain, more accessible, search friendly and mobile ready

Table 11: Final answers

9. Conclusion

The CampusConnect case study shows that website development rests on the client–server model, the separation of structure, presentation and behaviour, and a clear development life cycle. HTML5 supports this with a simple document structure, semantic elements that describe the purpose of every region of the page, and many built-in features: improved forms, native audio and video, canvas and SVG graphics, and APIs for storage and location. Rebuilding the site with these features made the code shorter, the content easier for browsers, search engines and assistive technology to understand, and the pages usable on all devices without plug-ins. HTML5 is therefore the standard foundation for modern web applications deployed on LAMP and WAMP servers.

10. Glossary of Key Terms

Term	Meaning
HTML5	Current version of HyperText Markup Language that defines page structure and APIs
Semantic element	Tag whose name describes the meaning of its content
DOCTYPE	Declaration that puts the browser in standards mode
DOM	Tree representation of the page that scripts can read and change
Viewport	Visible area of the page on a device screen
Canvas / SVG	Bitmap drawing area controlled by script / vector graphics in markup
Web Storage	Browser storage of key-value data (localStorage, sessionStorage)
Landmark	Region of a page identified for assistive technology (nav, main, aside)
W3C / WHATWG	Organisations that maintain the web standards

Table 12: Glossary